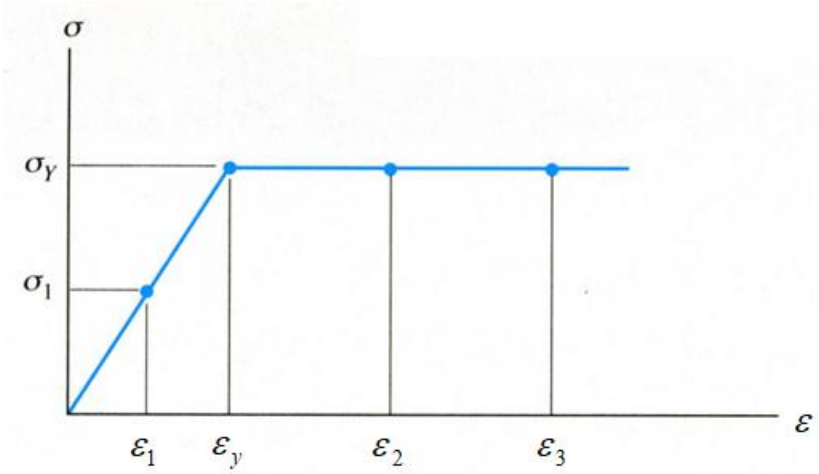
CH14　非線(彈)性分析

14-1.軸向負載之非線(彈)性分析

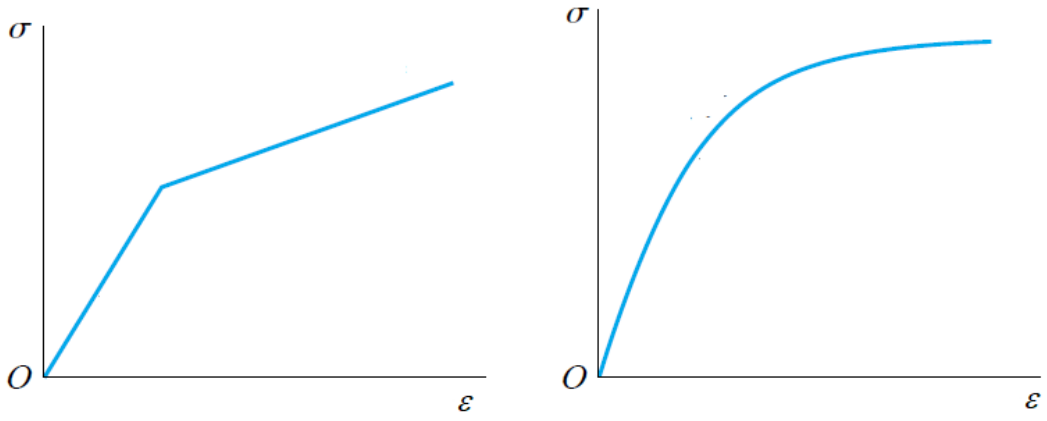
一、材料彈性-完全塑性 (elastic-perfectly plastic)

構件設計成負載後會導致材料降伏而產生永久變形。此種構件通常使用具有高延性的金屬如退火低碳鋼，其應力 - 應變曲線可模型化如圖所示。具有此種性質的材料通常稱為彈性-完全塑性 (elastic-perfectly plastic) 或彈塑 (elastoplastic)



二、非線性材料

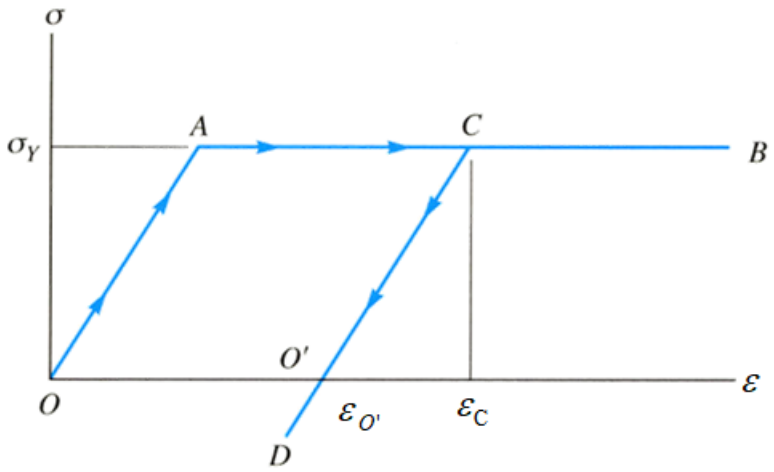
由兩段不同斜率的直線所組成之材料稱為雙線性材料，其應力-應變曲線可模型化如圖(a)所示，另，圖(b)所示為一般化非線性關係曲線，其中並無明顯的線性行為



(a)　(b)

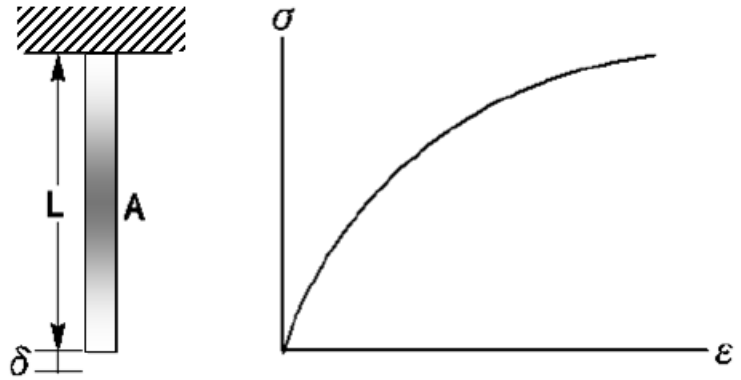
三、殘留應力 (residual stress)

若一軸向負載構件或一組構件為靜不定系統而可承受拉力或壓力，在受到過度負載時會導致材料降伏，當負載移除後會在構件產生殘留應力（residual stress)。產生殘留應力的理由是於卸載後材料的彈性恢復所致。欲解此種問題，可將構件的負載及卸載過程考慮成正值負載（ 加載 ）及負值負載（ 卸載 ）的重疊。在負載過程由 O 至 C，產生塑性應力分佈，而在卸載過程 CD，只有彈性應力分佈。



範例 1

圖中桿件材料(比重量為γ)之應力-應變關係為$\sigma = c\varepsilon^{1/2}$，試求桿子由於自重所生之伸長量。桿子長度為 L，橫斷面積為 A。

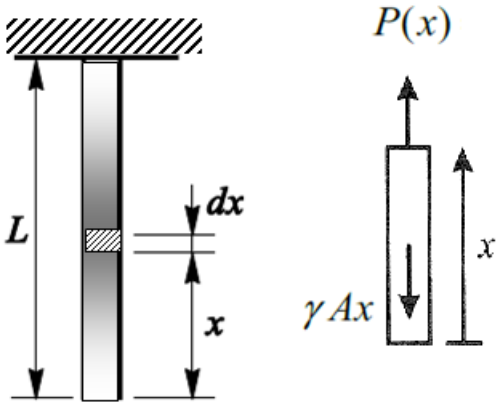


【中山機電】

【解析】

(a)內力分析

考慮距底部x處斷面由靜平衡方程式可得



$$P(x) = \gamma Ax \ ， \ \sigma(x) = \frac{P(x)}{A} = \gamma x$$

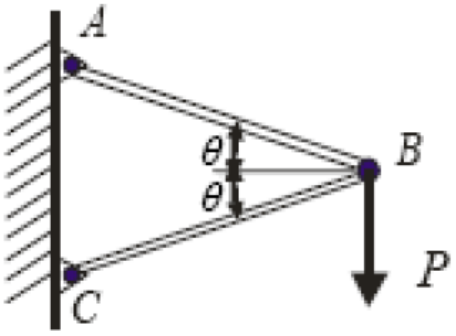
(b)應力與應變關係

$$\sigma = c\varepsilon^{1/2} \ ， \ \sigma^2 = c^2\varepsilon \ ， \ (\gamma x)^2 = c^2\varepsilon(x) \ ， \ \varepsilon(x) = \frac{\gamma^2 x^2}{c^2}$$

$$d\delta = \varepsilon(x)dx = \frac{\gamma^2}{c^2}x^2dx \ ， \ \delta = \int_0^L d\delta = \frac{\gamma^2}{c^2}\int_0^L x^2dx = \frac{\gamma^2 L^3}{3c^2}$$

範例 2

如圖所示，完全相同之桿件 AB、BC(長度為 L，斷面積為 A)，端點均為拼接，應力 σ 與應變 ε 滿足 $\sigma = E\varepsilon^3$(E 為常數)之非線性材料，於 B 點承受垂直作用力 P，不計桿件重量。試求：(一)B 點之垂直位移 。(二)構架之總應變能 U。(三)卡式定理 $\delta_B = \partial U / \partial P$ 是否滿足？請說明原因。(95 土木高考三等)



【解析】

(一)

1. 應變能密度 $u = \int_o^\varepsilon \sigma d\varepsilon = E\int_o^\varepsilon \varepsilon^3 d\varepsilon = \frac{E}{4}\varepsilon^4 = \frac{\sigma^{\frac{4}{3}}}{4E^{\frac{1}{3}}}$ ：

又桿件 AB、BC 長度為 L，斷面積為 A，內力為 N

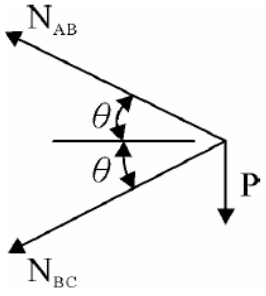
應變能 $U = \frac{(\frac{N}{A})^{\frac{4}{3}} \times AL}{4E^{\frac{1}{3}}} = \frac{(N)^{\frac{4}{3}}L}{4(AE)^{\frac{1}{3}}}$

同理輔能密度 $u^* = \int_o^\sigma \varepsilon d\sigma = \int_o^\sigma (\frac{\sigma}{E})^{\frac{1}{3}} d\sigma = \frac{3\sigma^{\frac{4}{3}}}{4E^{\frac{1}{3}}}$

輔能 $U^* = \frac{3(\frac{N}{A})^{\frac{4}{3}}}{4E^{\frac{1}{3}}}(AL) = \frac{3(N)^{\frac{4}{3}}L}{4(AE)^{\frac{1}{3}}}$

(二)

2. 取 AB 及 BC 受力自由體圖：



$N_{AB} = \frac{P}{2\sin\theta}$ (拉)，$N_{BC} = -\frac{P}{2\sin\theta}$ (壓)

總應變能 $U = 2[\frac{L}{4(AE)^{\frac{1}{3}}}(\frac{P}{2\sin\theta})^{\frac{4}{3}}] = \frac{P^{\frac{4}{3}}L}{4(2AE)^{\frac{1}{3}}(\sin\theta)^{\frac{4}{3}}}$

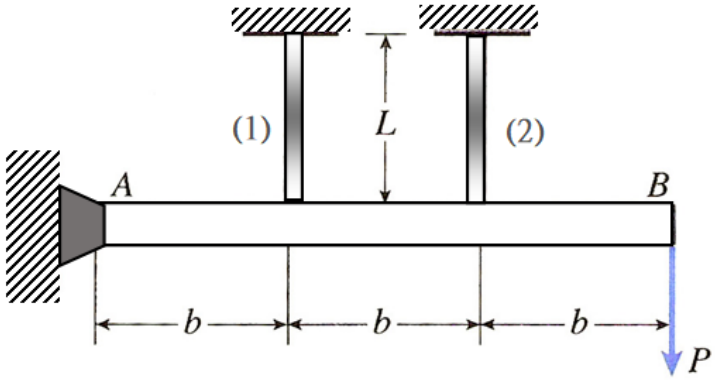
總輔能 $U^* = 2[\frac{3L}{4(AE)^{\frac{1}{3}}}(\frac{P}{2\sin\theta})^{\frac{4}{3}}] = \frac{3P^{\frac{4}{3}}L}{4(2AE)^{\frac{1}{3}}(\sin\theta)^{\frac{4}{3}}}$

3. 由卡式第一定理：$\delta_B = \frac{\partial U^*}{P} = \frac{P^{\frac{1}{3}}L}{(2AE)^{\frac{1}{3}}(\sin\theta)^{\frac{4}{3}}}$ (↓)。

4. 卡式第二定理僅可用於線彈性系統：$\delta_B = \frac{\partial U}{\partial P}$ 並不成立。

範例 3

The structure shown in Fig. consists of a horizontal beam AB (assumed to be rigid) supported by two identical bars (bars 1 and 2) made of an elastoplastic material. The bars have length L and cross-sectional area A, and the material has yield stress σ_y, yield strain ε_y, and modulus of elasticity $E=\frac{\sigma_y}{\varepsilon_y}$. The beam has length 3b and supports a load P at end B. (a) Determine the yield load P_y and the corresponding yield displacement δ_y at the end of the bar (point B). (b) Determine the plastic load P_P and the corresponding plastic displacement δ_P at point B. (c) Construct a load-displacement diagram relating the load P to to the displacement δ_B of point B

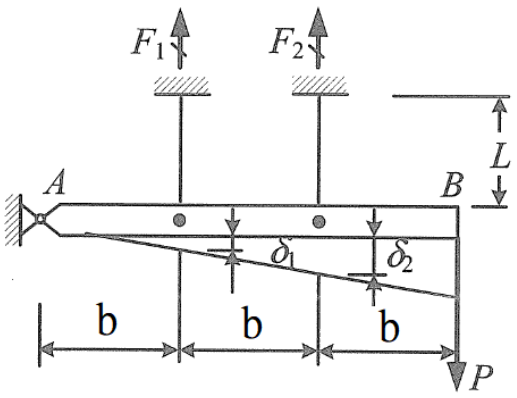


【中文翻譯】

示於圖 的結構包含了一支水平樑 AB （ 假設是剛性 的 ），且由一種彈塑性材料所製之兩支相同的桿 （ 桿 1 及 2) 所支 撐。桿長 L ，且橫截面面積爲 A ，且材料有降伏應力 σ_y 、降伏應變 ε_y 及彈性模數 $E=\frac{\sigma_y}{\varepsilon_y}$ 。樑長 $3b$ ，並在端 B 支撐負載 P 。(a) 求降伏負載 P_y 及在點 B 之相對應的降伏位移 δ_y ；(b) 求塑性 負載 P_p 及在 B 點相對應的塑性 位移 δ_p ；(c) 建構一負載 - 位移圖， 以關連負載 P 到點 B 之位移 δ_B 。

【解析】

(a)內力分析



(1) 靜力平衡式

$\sum M_A = 0$　　$b F_1 + 2b F_2 = 3P\ b$ 即 $F_1 + 2F_2 = 3P$ .....①

(2) 變位一致性：$\delta_2 = 2\delta_1$ .....②

(3) 力與變形關係式

$$\delta_1 = \frac{F_1 L}{EA} \quad \text{.....③}$$

$$\delta_2 = \frac{F_2 L}{EA} \quad \text{.....④}$$

由②～④三式聯立求解，即得 $2F_1 = F_2$

由②～④三式聯立求解，即得 $2F_1 = F_2$

再代入①式得 $F_1 = \frac{3P}{5}$ ； $F_2 = \frac{6P}{5}$

桿 2 具有較大的力會是最早達到降伏應力者 ，桿 2 之力會 是 $F_2 = \sigma_y A$

$$P_y = \frac{5\sigma_y A}{6}$$

桿 2 相對應的伸長量 爲 $\delta_2 = \sigma_y L / E$，在點 B 的降 伏位移爲

$$\delta_y = \frac{3\delta_2}{2} = \frac{3\sigma_y L}{2E}$$

(b)

當極限負載 P_u 達到時， $F_1 = F_2 = \sigma_Y A$ ，故由①式得

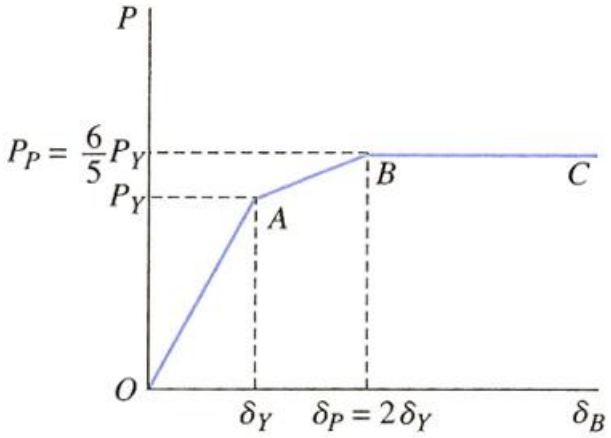
$P_P = \sigma_Y A$

在此負載下，左邊電線剛達降伏應力，故其伸長量

$\delta_1 = \frac{\sigma_Y h}{E}$ 。

而此時在 B 端之撓度爲 $\delta_P = 3\delta_1 = \frac{3\sigma_Y h}{E}$

(c)


P
$P_P = \frac{6}{5}P_Y$
P_Y
A
B
C
O
δ_Y
$\delta_P = 2\delta_Y$
δ_B

14-2.扭轉負載下之非線(彈)性分析

一、 扭轉負載下之非線(彈)性分析

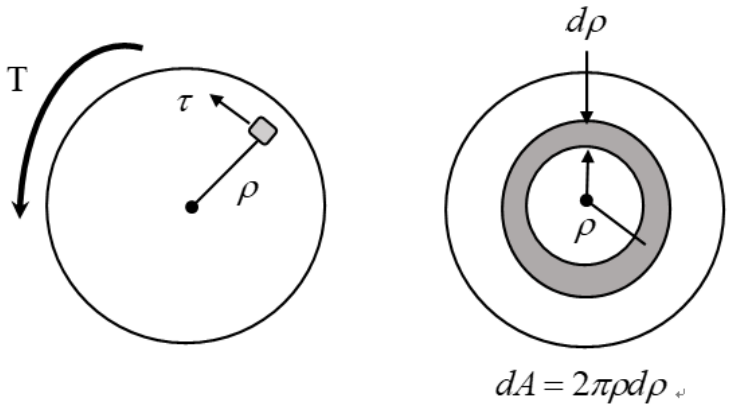
(一)桿件扭力

若扭轉負載過載，材料將降伏，也因此塑性分析必須用來計算剪應力分佈及扭轉角，為了執行此分析，必須同時滿足軸的變形及平衡條件，則其扭矩為:

$$T = \int \rho \tau dA$$

τ 所作用的面積 dA 定義為微分環，面積為 $dA = 2\pi\rho d\rho$，如圖所示，則上列方程式可寫成

$$T = \int \rho \tau 2\pi\rho d\rho = 2\pi \int \tau \rho^2 d\rho$$



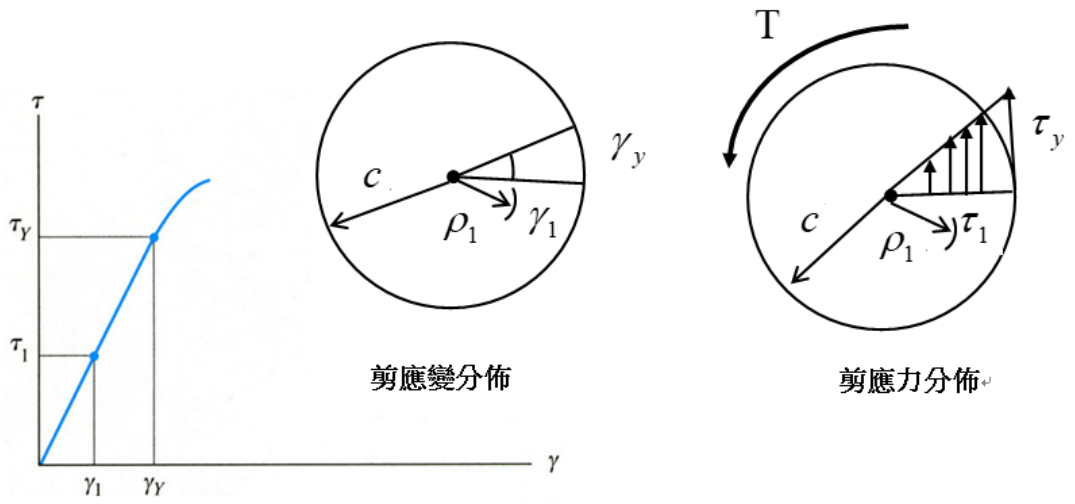
(二) 最大彈性扭矩

若扭轉使軸的外表面產生最大彈性剪應變 γ_Y，則沿徑向線的剪應變分佈看起來將如圖所示，最大彈性扭矩式求得:

$$T_y = 2\pi \int \tau \rho^2 d\rho = 2\pi \int \tau_y (\frac{\rho}{c}) \rho^2 d\rho$$

$$\Rightarrow T = \frac{\pi}{2} \tau_y c^3$$

$$d\phi = \gamma \frac{dx}{\rho}$$



(三)**彈-塑性扭矩**

若所施加的力矩造成材料超過彈性極限，則應力分佈與軸心算起的徑向距離無正比關係，而是利用剪應力-剪應變圖與平衡式，將力矩關聯於應力分佈，若扭轉負載過載材料將降伏，當剪應變增加至γ_y，降伏會首先在其外表面上，扭矩負載持續增加，則降伏的邊界會向軸中心移動，當$\rho = c$時，如圖所示會產生一彈性核心(elastic core)，對於彈塑材料的分析如下:

$$T = 2\pi \int_0^c \tau \rho^2 d\rho$$

$$= 2\pi \int_0^{\rho_Y} \left(\tau_Y \frac{\rho}{\rho_Y} \right) \rho^2 d\rho + 2\pi \int_{\rho_Y}^c \tau_Y \rho^2 d\rho$$

$$= \frac{2\pi}{\rho_Y} \tau_Y \int_0^{\rho_Y} \rho^3 d\rho + 2\pi \tau_Y \int_{\rho_Y}^c \rho^2 d\rho$$

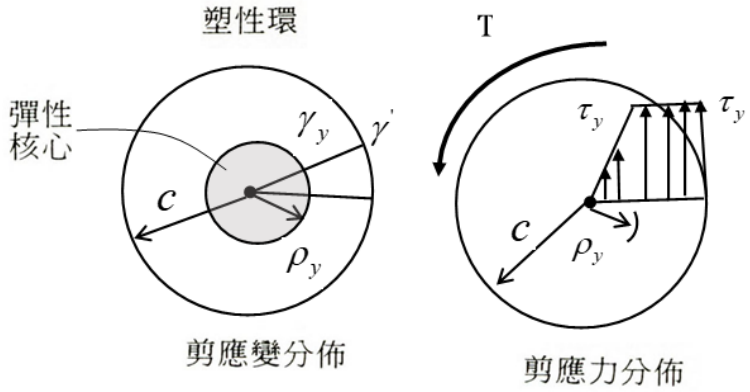
$$= \frac{\pi}{2\rho_Y} \tau_Y \rho_Y^4 + \frac{2\pi}{3} \tau_Y (c^3 - \rho_Y^3)$$

$$= \frac{\pi \tau_Y}{6} (4c^3 - \rho_Y^3)$$

若扭矩 T 再增加使得彈性核心的半徑縮小直至所有材料降伏，即$\rho \rightarrow 0$，則塑性扭矩 Tp 代表軸可承受的最大扭矩:

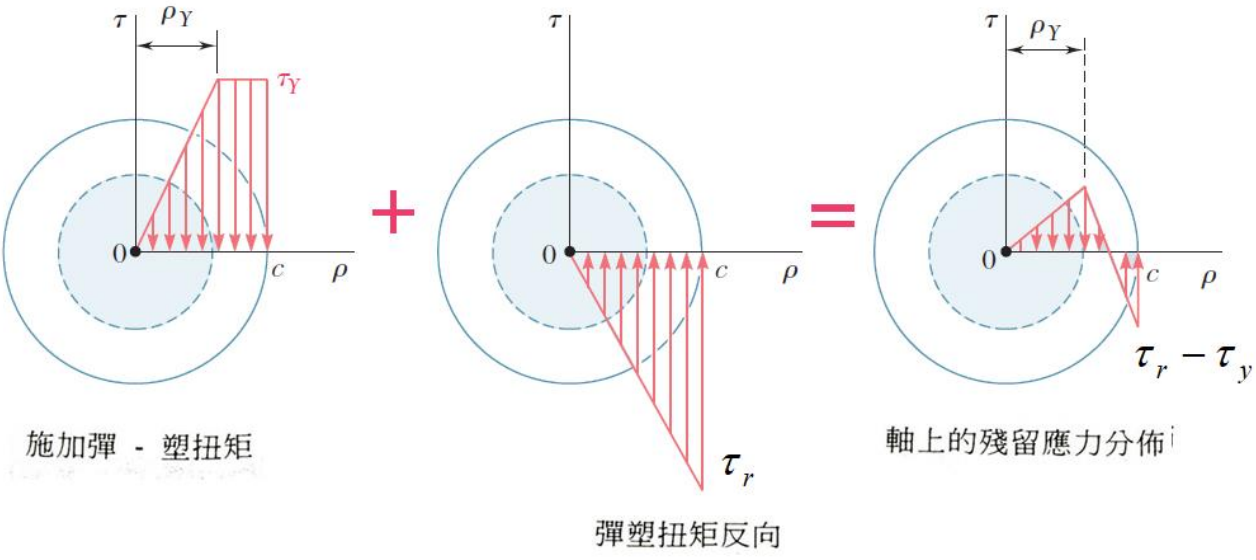
$$T_p = 2\pi \int_0^c \tau_Y \rho^2 d\rho$$

$$= \frac{2\pi}{3} \tau_Y c^3$$



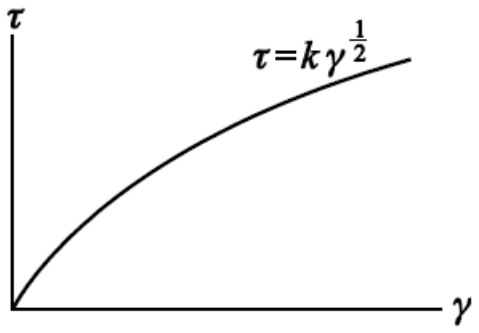
二、 扭轉負載下之殘留應力

當軸受到扭轉而產生塑性剪應變時，移去扭矩後會有些剪應力殘留在軸上。此應力稱為殘留應力，其分佈情況可由重疊原理及彈性恢復求得。



範例 4

半徑為 R 之圓桿，其剪應力與剪應變關係如圖所示，當此圓桿承受扭矩 T 作用時，試求(a)圖桿之扭轉角(以 T、R、k 及 L 表示)(b)圓桿之最大剪應力(以 T、R 表示)



【解析】

(a)計算扭轉角

$$T=\int_A \rho(\tau dA)=\int_0^R \rho(k\gamma^{1/2})2\pi\rho d\rho=\int_0^R \rho k\left(\frac{\rho\theta}{L}\right)^{1/2}2\pi\rho d\rho$$

$$=2\pi k\left(\frac{\theta}{L}\right)^{1/2}\int_0^R \rho^{5/2}d\rho=\frac{4\pi k}{7}\left(\frac{\theta}{L}\right)^{1/2}R^{7/2}$$

將上式整理得 $\theta=\dfrac{49T^2L}{16\pi^2k^2R^7}$

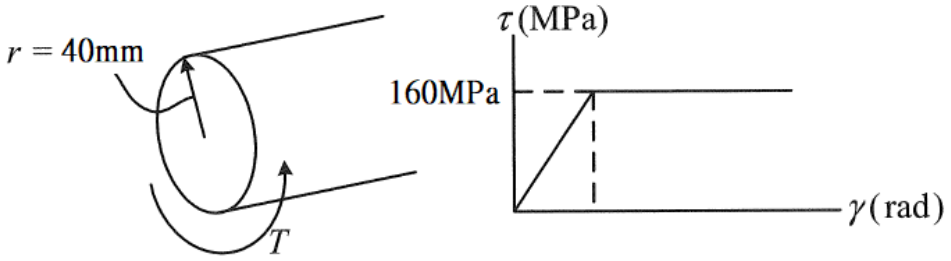
(b)應力分析

$$\gamma_{\max}=\frac{R\theta}{L}=\frac{49T^2}{16\pi^2k^2R^6}$$

$$\tau_{\max}=k\gamma_{\max}^{1/2}=k\left(\frac{49T^2}{16\pi^2k^2R^6}\right)^{1/2}=\frac{7T}{4\pi R^3}$$

範例 5

The solid shaft is made of an elastic-perfectly plastic material as shown. Determine the torque T needed to form an elastic core in the shaft having a radius of ρ_y = 20mm .If the shaft is 3 m long, through what angle does one end of the shaft twist with respect to the other end? When the torque is removed, determine the residual stress distribution in the shaft and the permanent angle of twist



【成大機械】

【中文翻譯】

一實心圓桿長度為3m半徑為40mm之，用彈塑材料製造(τ_y = 160 MPa 、G = 40 GPa)而成。(a)試求使圓桿產生ρ_y = 20mm之彈性核心半徑所需之扭矩，及此時圓桿之扭轉角，(b)將(a)中之扭矩移除，試求殘留應力之分佈及永久扭轉角。

【解析】

(一)扭矩分析

$$T = 2\pi\int_0^c \tau\rho^2 d\rho$$

$$= 2\pi\int_0^{\rho_Y}\left(\tau_Y\frac{\rho}{\rho_Y}\right)\rho^2 d\rho + 2\pi\int_{\rho_Y}^c \tau_Y\rho^2 d\rho$$

$$= \frac{2\pi}{\rho_Y}\tau_Y\int_0^{\rho_Y}\rho^3 d\rho + 2\pi\tau_Y\int_{\rho_Y}^c \rho^2 d\rho$$

$$= \frac{\pi}{2\rho_Y}\tau_Y\rho_Y^4 + \frac{2\pi}{3}\tau_Y(c^3-\rho_Y^3)$$

$$= \frac{\pi\tau_Y}{6}(4c^3-\rho_Y^3)$$

$$T = \frac{\pi \tau_Y}{6}\left(4c^3 - \rho_Y^3\right)$$

$$= \frac{\pi(160)(10^6)}{6}\left[4\left(0.04^3\right) - 0.02^3\right]$$

$$= 20776.40\ \text{N}\cdot\text{m} = 20.8\ \text{kN}\cdot\text{m}$$

$$\phi = \frac{\gamma_Y}{\rho_Y}L = \left(\frac{0.004}{0.02}\right)(3) = 0.600\ \text{rad} = 34.4^\circ$$

(二)將扭矩移除

彈性回復之扭轉角ϕ'及剪應力τ'

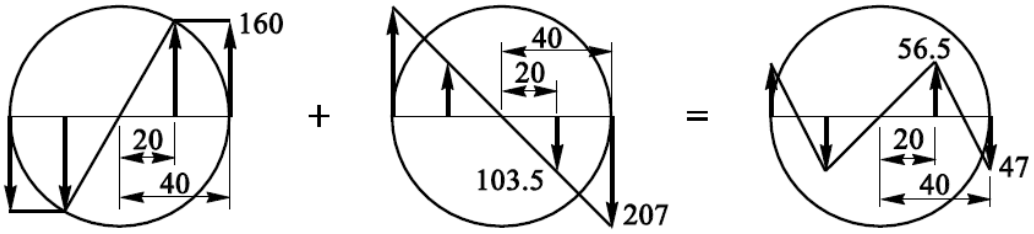
$$\phi' = \frac{T'L}{GJ} = \frac{(20.8\times10^3)(3000)}{(40)\left(\frac{\pi\times80^4}{32}\right)} = 0.388\ \text{rad} = 22.2^\circ$$

永久扭轉角為$\phi_R = \phi - \phi' = 34.4^\circ - 22.2^\circ = 12.2^\circ$

$$\tau_{\max}' = \frac{16T'}{\pi d^3} = \frac{16(20.8\times10^6)}{\pi(80)^3} = 207\ \text{MPa}$$

$\rho_y = 20$ mm 處，$\tau' = \frac{\rho_y}{r}\tau_{\max}' = \frac{20}{40}(207) = 103.5$ MPa

斷面剪應力分佈如圖所示



14-3.彎矩負載下之非線(彈)性分析

一、 非彈性彎曲

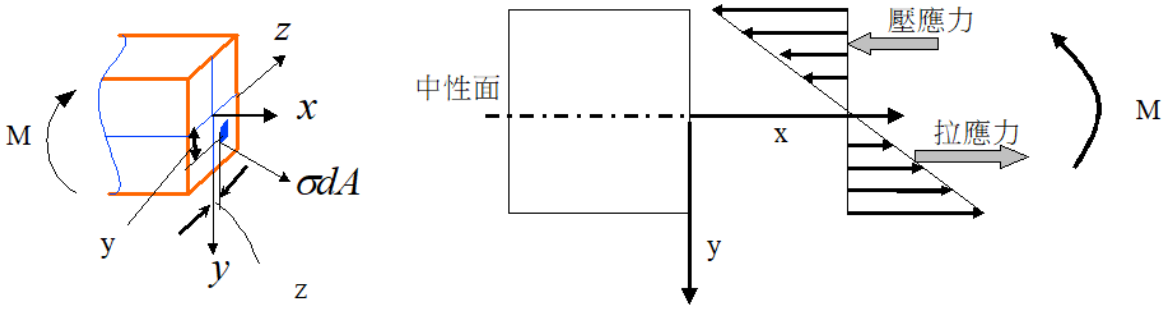
線性正向應變分佈，基於幾何考慮，在前節中已描述，材料內所產生正向應變乃從截面中性軸零線性變化至距中性軸最遠之點最大，因只有一合內彎矩（resultant internal moment）作用在截面上，故由應力分佈產生之合力必須等於零。

1. 如圖橫截面上座標爲(y,z)的點的正應力爲σ，截面上各點的微內力σdA組成與橫截面垂直的空間平行力系。

$$\sum \mathrm{F}_x = 0 \Rightarrow N = \int_A \sigma dA = 0 \text{(平行 } x \text{ 軸的軸力)}$$

$$\mathrm{M} = \int y\sigma dA$$

2. 繪出構件截面上應力分佈圖，利用$\sigma - \varepsilon$曲線定出相對應於應變之應力值。
3. 樑之截面上的正向應變分佈只與幾何形狀有關，並且證實為線性與施加負載無關，然而正向應力的分佈一旦應變分佈得知後，則須由材料性質，或應力 / 應變關係圖來求得。
4. 中性軸的位置可由截面上之合力為零的條件來求得。

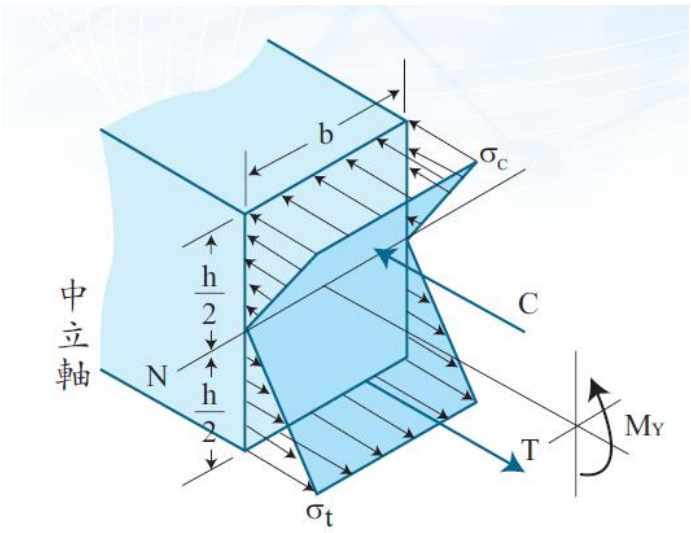


二、 塑性彎矩

(一)降伏彎曲應力

1. 對一已知力矩 M，假設中性軸位置及“線性”應變分佈形狀。
2. 繪出構件截面上應力分佈圖，求出由拉伸及壓縮應力區所包圍體積，因在截面上各區乃表示合拉 T 及合壓力 C，得知兩區間體積相同。若兩力不等，則中性軸位置須作調整（ 零應變點 ）。
3. 如下圖所示，若降伏彎矩 My 作用下使構件正好降伏，各力作用線通過其相關三角應力塊之形心，得

$$\mathrm{M}_y = \int y\sigma dA = C \times (\frac{2}{3}) \times \frac{h}{2} + T \times (\frac{h}{2}) = 2 \times [\frac{1}{4} bh\sigma_y \times (\frac{2}{3}) \times \frac{h}{2}] = \frac{1}{6} bh^2 \sigma_y$$



(二) **塑性彎矩分析**

截面上之內彎矩合須等於其應力分佈對中性軸所產生的彎矩如圖所示

$$\sigma_x = -\frac{\sigma_y}{y_Y} y$$

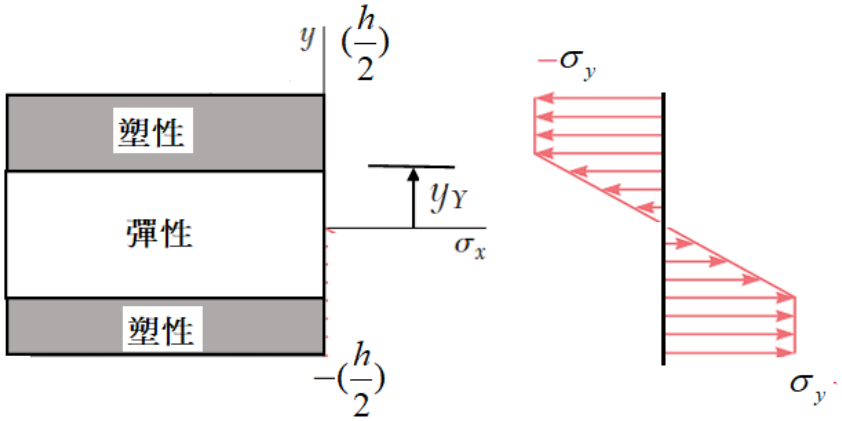
$$\mathrm{M} = -2b\int_0^{y_Y} y(-\frac{\sigma_y}{y_Y} y)dy - 2b\int_{y_Y}^{\frac{h}{2}} y(-\sigma_y)dy$$

$$\Rightarrow \mathrm{M} = \frac{2}{3} by_Y^2 + b(\frac{h}{2})^2\sigma_y - by_Y^2\sigma_y$$

$$\Rightarrow \mathrm{M} = b(\frac{h}{2})^2\sigma_y[1-\frac{4y_Y^2}{3(h)^2}]$$

$$\Rightarrow \mathrm{M} = \frac{3}{2} M_y[1-\frac{4y_Y^2}{3(h)^2}]$$

其中 σ_y :降伏應力



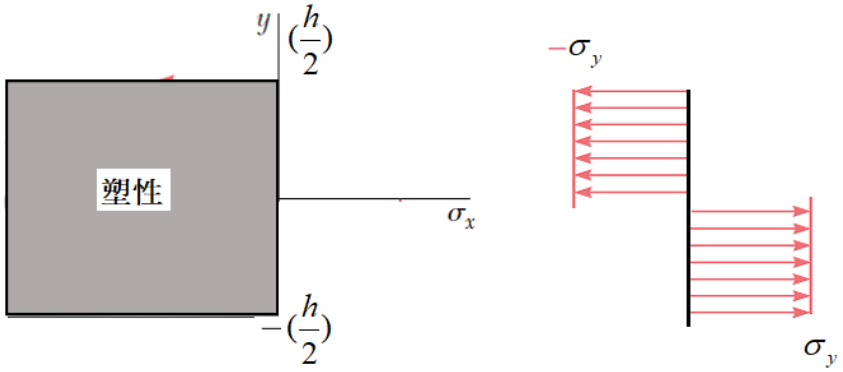
(三)完全塑性

完全塑性性質是假設其正向應力在截面上的分佈為一定值，在彎矩沒有增加的情

況下，樑仍會繼續彎曲，此彎矩稱之為塑性彎矩。

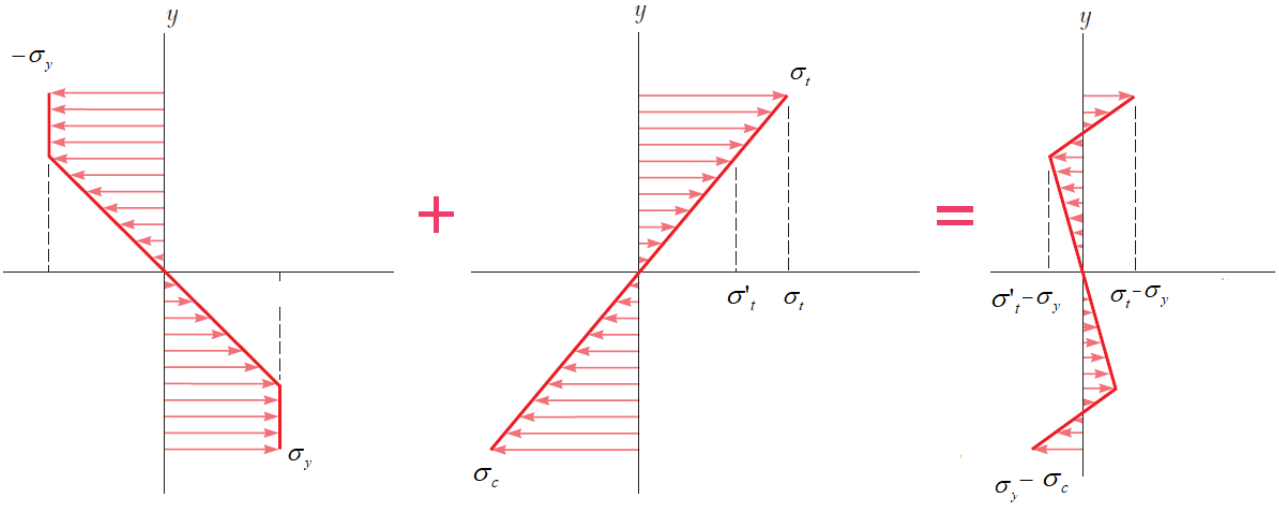
$$M_P = -2b\int_0^{\frac{h}{2}} y(-\sigma_y)dy = \frac{1}{4}bh^2\sigma_y$$

$$\Rightarrow M_P = \frac{3}{2}M_y$$



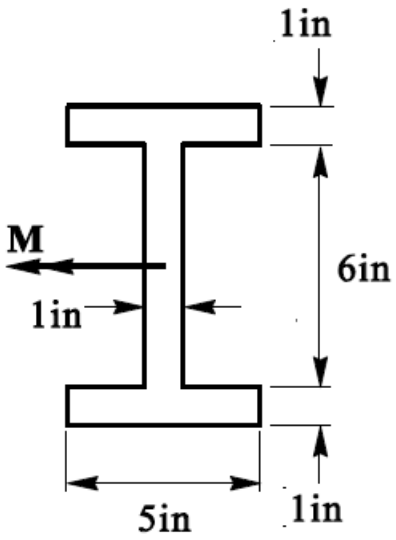
三、殘留應力

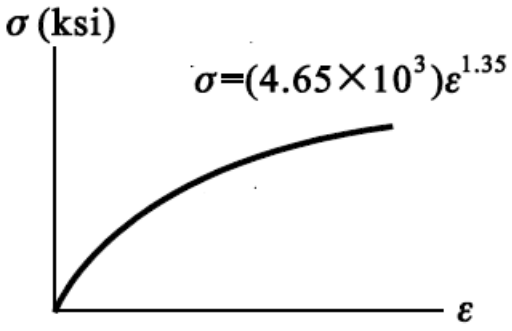
若一樑受承負載使得材料降伏，則移去負載將使材料內產生殘留應力 (residual stress)，當考慮疲乏 (fatigue) 及其他機械性質，殘留應力通常是很重要的，如同扭轉 (torsion) 吾人可應用重疊原理 (principles of superposition) 及彈性回復 (elastic recovery) 求得殘留應力分佈。



範例 6

圖中寬翼型斷面之梁由塑膠材料製造，其應力與應變之關係為$\sigma = (4.65\times10^3)\varepsilon^{1.35}$ ksi。設梁內容許之最大應變為$\varepsilon_{max} = 0.005$，試求梁所能承受之最大彎矩。





【解析】

(一)應力分析

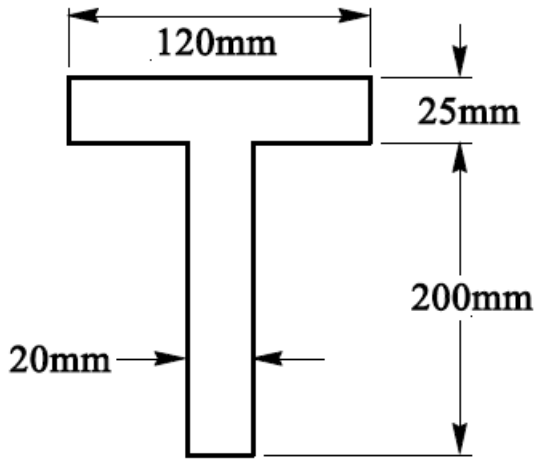
$$P=\frac{P_{max}}{c}y=\frac{0.005}{4}y$$

$$\sigma=(4.65\times10^3)\left(\frac{0.005}{4}y\right)^{1.35}=0.560y^{1.35}$$

$$M=2\left[\int_0^3 y(0.560y^{1.35})(1dy)+\int_3^4 y(0.560y^{1.35})(5dy)\right]=120.8\ \text{kip-in}$$

範例 7

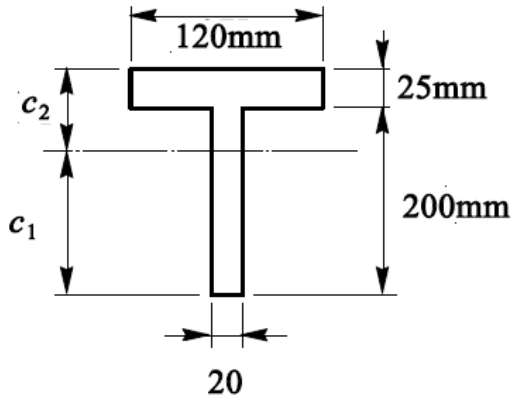
試求圖中 T 型斷面之形狀因數 f。



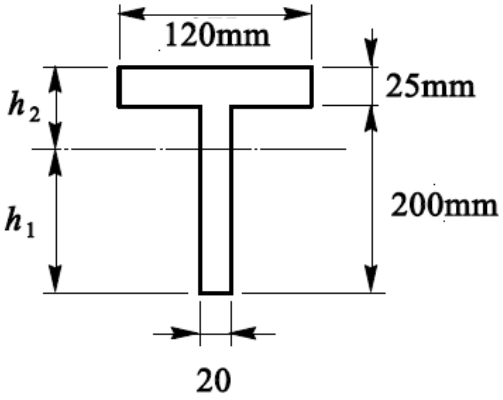
【交大土木】

【解析】

(一)彈性斷面分析



(彈性彎曲)



(塑性彎曲)

面積：$A = (20 \times 200) + (120 \times 25) = 7000 \text{ mm}^2$

$$c_2 = \frac{(200\times 20)(125) + (120\times 25)(12.5)}{7000} = 76.8 \text{ mm}\ ,\ c_1 = 225 - c_2 = 148.2 \text{ mm}$$

$$I = \frac{1}{3}(20)(200)^3 + \frac{1}{3}(120)(25)^3 - (7000)(76.8-25)^2 = 35.18\times 10^6 \text{ mm}^4$$

$$S = \frac{I}{c_1} = \frac{35.18\times 10^6}{148.2} = 237.4\times 10^3 \text{ mm}^3$$

(一)塑性斷面分析

塑性中立軸位置：

$$20h_1 = \frac{A}{2} = 3500 \text{，} h_1 = 175 \text{ mm} \text{，} h_2 = 225 - 175 = 50 \text{ mm}$$

$$\overline{y_1} = \frac{h_1}{2} = 87.5 \text{ mm}$$

$$\overline{y_2} = \frac{(120 \times 25)(50 - 12.5) + (20 \times 25)(12.5)}{3500} = 33.93 \text{ mm}$$

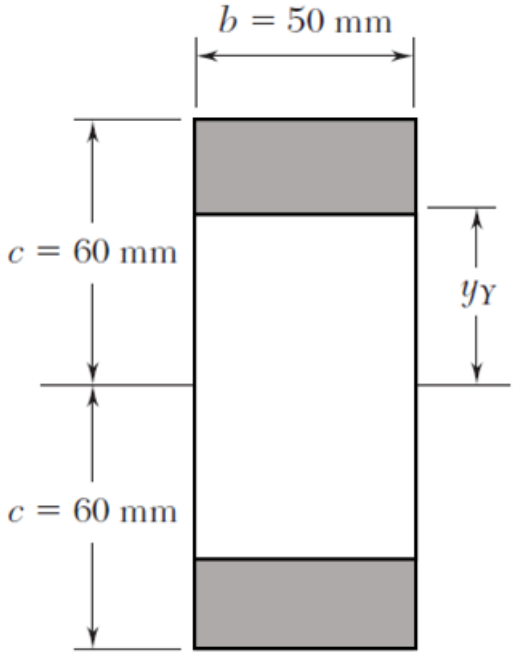
$$Z = \frac{A}{2}(\overline{y_1} + \overline{y_2}) = (3500)(87.5 + 33.93) = 425 \times 10^3 \text{ mm}^3$$

$$f = \frac{Z}{S} = \frac{425}{237.4} = 1.79$$

範例 8

A member of uniform rectangular cross section $50\times120mm$ (Fig.) is subjected to a bending moment $M=36.8KN-m$. Assuming that the member is made of an elastoplastic material with a yield strength of 240 MPa and a modulus of elasticity of 200 GPa, determine (a) the thickness of the elastic core, (b) the radius of curvature of the neutral surface.

If the bending moment has been decreased from its maximum value of 36.8 kN -m back to zero, determine (C) the distribution of the residual stresses, (d) the radius of curvature,



【105 台大機械】

【中文翻譯】

如圖所示一均勻矩形橫截面$50\times120mm$的構件受到彎矩，假設構件由降服強度240MPa，彈性模數為200GPa的彈塑性材料製成，試求出（a）彈性核心的厚度，（b）中性面的曲率半徑。 若彎曲力矩從其最大值36.8(kN-m)下降到0後試求（C）殘餘應力的分佈;（d）曲率半徑;

【解析】

(a)彈性區厚度

(一)斷面分析

$$S=\frac{1}{6}bh^2=\frac{1}{6}\times50\times120^2=120000(mm^3)$$

(二) 降伏彎矩

$$M_y = \frac{1}{6}bh^2\sigma_y = 120000\times 240 = 28800000(N-mm) = 28.8(KN-m)$$

$$M = -2b\int_0^{y_Y} y(-\frac{\sigma_y}{y_Y}y)dy - 2b\int_{y_Y}^{\frac{h}{2}} y(-\sigma_y)dy$$

$$\Rightarrow M = \frac{2}{3}by_Y^2 + b(\frac{h}{2})^2\sigma_y - by_Y^2\sigma_y$$

$$\Rightarrow M = b(\frac{h}{2})^2\sigma_y[1-\frac{4y_Y^2}{3(h)^2}]$$

$$\Rightarrow M = \frac{3}{2}M_y[1-\frac{4y_Y^2}{3(h)^2}] \quad \Rightarrow 36.8 = \frac{3}{2}\times 28.8\times[1-\frac{4y_Y^2}{3(120)^2}] \Rightarrow y_Y = 40(mm)$$

(b)曲率半徑

$$\varepsilon_y = \frac{\sigma_y}{E} = \frac{240}{200\times 10^3} = 1.2\times 10^{-3}$$

$$\rho = \frac{y_Y}{\varepsilon_y} = \frac{40}{1.2\times 10^{-3}} = 33333.33(mm)$$

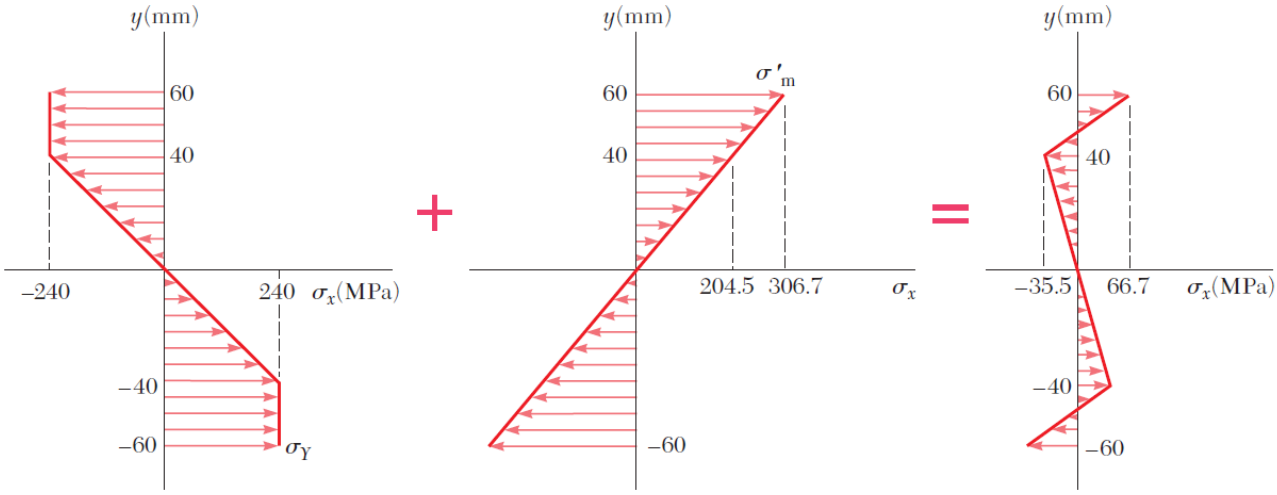
(c)殘留應力

彎矩移去後產生彈性回復，相當於施加一反向彎矩M' = 36.8 kN-m 抵銷原施加之彎矩

$$\sigma'_m = \frac{My}{I} = \frac{M}{S} = \frac{36.8\times 10^6}{120000} = 306.7(MPa)$$

殘留應力：y = 40 mm 處，$\sigma_R = (-240) + 204.4 = -35.5$ MPa

y = 60 mm 處，$\sigma_R = (-240) + 306.7 = 66.7$ MPa



(d)曲率半徑

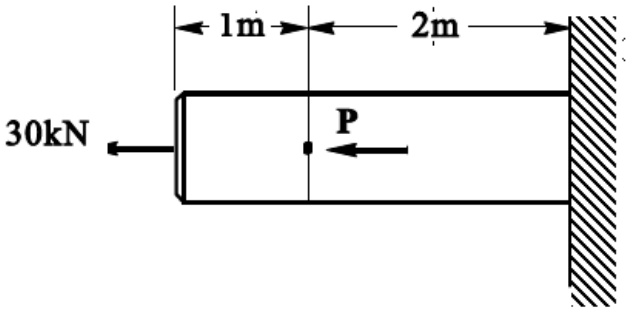
$$\varepsilon_x = \frac{\sigma_x}{E} = \frac{-35.5}{200\times 10^3} = -177.5\times 10^{-6}$$

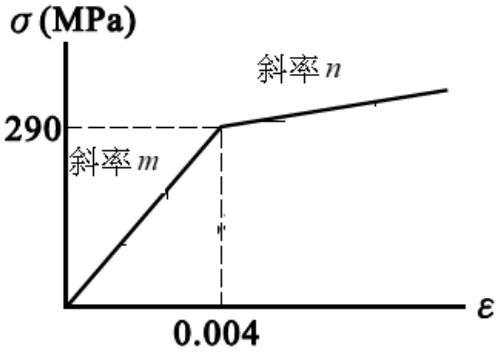
$$\rho = \frac{y_Y}{\varepsilon_y} = \frac{40}{177.5 \times 10^{-6}} = 225352.11(mm)$$

實戰演練

例 1

圖中直徑 20mm 之鋁合金桿子 承受兩個軸力 P 及 30 kN，已知鋁合 金之應力-應變圖如圖所示，其中斜率 $m=\frac{1}{72500}$、斜率 $n=\frac{1}{9700}$，試求下列兩種情況桿子之伸長量，(a)P =20 kN，(b)P = 65 kN。





【解析】

(c)內力分析

$$P=20\text{ kN}\cdot F_1=50\text{ kN}\cdot F_2=30\text{ kN}\cdot A=\frac{\pi}{4}(20)^2=314\text{ mm}^2$$

應力與應變分析

$$\sigma_1=\frac{50\times10^3}{314}=159\text{ MPa}\cdot\ \sigma_2=\frac{30\times10^3}{314}=95.5\text{ MPa}$$

$$\varepsilon_1=\frac{159}{72500}=2.193\times10^{-3}\ \cdot\ \varepsilon_2=\frac{95.5}{72500}=1.318\times10^{-3}$$

$$\delta=\varepsilon_1L_1+\varepsilon_2L_2=(2.193\times10^{-3})(2000)+(1.318\times10^{-3})(1000)=5.71\text{ mm}$$

(d)內力分析

$$P=65\text{ kN}\cdot F_1=95\text{ kN}\cdot F_2=30\text{ kN}$$

應力與應變關係

$$\sigma_1 = \frac{95\times10^3}{314} = 302.5\ \text{MPa}\ ,\ \sigma_2 = 95.5\ \text{MPa}\ ,\ \varepsilon_2 = 1.318\times10^{-3}$$

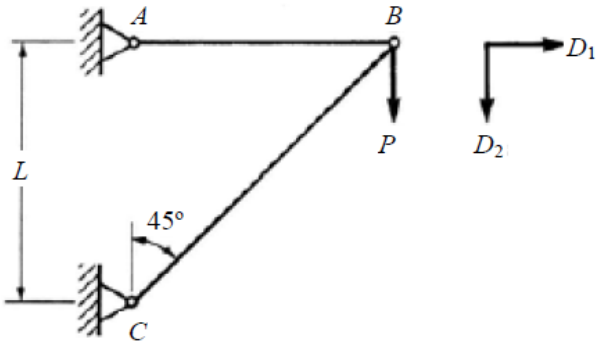
由內插法

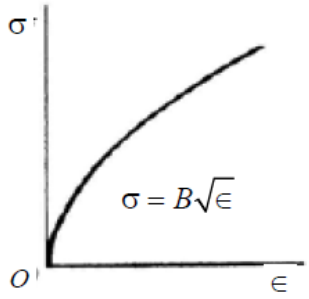
$$\frac{302.5-290}{\varepsilon_1 - 0.004} = 9700\ ,\ \varepsilon_1 = 5.29\times10^{-3}$$

$$\delta = \varepsilon_1 L_1 + \varepsilon_2 L_2 = (5.29\times10^{-3})(2000) + (1.318\times10^{-3})(1000) = 11.9\ \text{mm}$$

例 2

如圖所示之桁架(truss)，桿件 AB 和 BC 之斷面積為 A，其材料之應力和應變圖為 $\sigma = B\sqrt{\varepsilon}$ ，B 為常數，試求圖中 B 點之垂直位移 D_2 為何？(93 機械地方特考三等)





【解析】

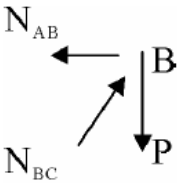
1. **由於材料為非彈性體，應用輔能觀念：**

$$u^* = \int \varepsilon d\sigma = \int_o^\sigma (\frac{\sigma}{B})^2 d\sigma = \frac{\sigma^3}{3B^2} = \frac{(\frac{N}{A})^3}{3B^2}$$

若桿件長度為 L，則輔應變能

$$U^* = u^* AL = \frac{N^3 L}{3B^2 A^2}$$

2. **取 B 節點自由體圖：**



$N_{BC} = -\sqrt{2}P$ (壓)， $N_{AB} = P$(拉)

$$U^* = \frac{P^3 L}{3B^2 A^2} + \frac{|-\sqrt{2}P|^3 \times \sqrt{2}L}{3B^2 A^2} = \frac{5P^3 L}{3B^2 A^2}$$

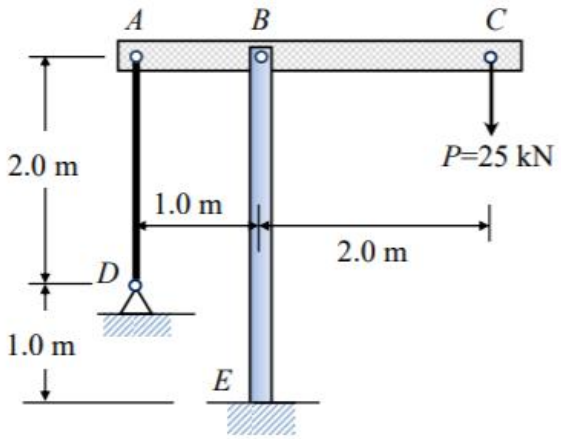
由卡氏第一定理 $\delta_B = \frac{\partial U^*}{\partial P} = \frac{5P^2 b}{B^2 A^2}$

例 3

如圖一(a)所示 ABC 為一剛體梁桿件（rigid beam）由桿件 BE 及金屬纜線 AD 支撐。假設 BE 桿件長 3.0 m、斷面積為 2500 mm^2 且彈性係數 E=200 GPa；AD 纜線之長為 2.0 m、斷面積為 140 mm^2、降伏應力（yielding stress）σ_y=255 MPa 且其應力應變關係如下式及圖一(b)所示：

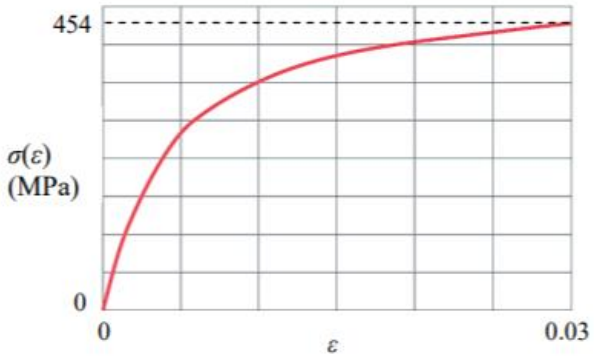
$$\sigma(\varepsilon)=\frac{124000\varepsilon}{1+240\varepsilon} \qquad 0\le\varepsilon\le 0.03 \qquad (\sigma\text{ 之單位為MPa})$$

㈠請問金屬纜線 AD 之彈性係數 E(GPa)？（5 分）
㈡試計算 C 點之垂直位移？（10 分）
㈢試計算當作用於 C 點之外力移除後，纜線 AD 之永久變形量（permanent set）？（10 分）



(a)

圖一



(b)

【102 結構技師】

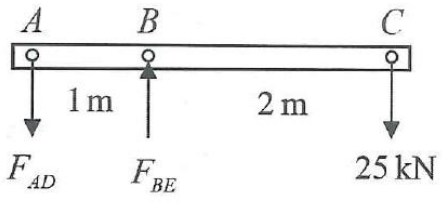
【解析】

（1） 彈性模數

由提議可知 $\sigma=\dfrac{124000\varepsilon}{1+240\varepsilon}$

$$E=\frac{\sigma}{\varepsilon}=\frac{124000}{1+240\varepsilon}$$

（2） 內力分析



$\Sigma M_B=0$ $\quad F_{AD}\times 1=25\times 2$

$F_{AD}=50\text{ kN}$

$\Sigma F_y=0$ $\quad F_{BE}=75\text{ kN}$

（3） 應力應變分析

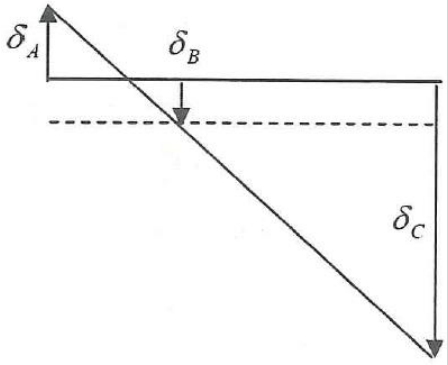
$$\delta_B = \frac{(75)(3000)}{(200)(2500)}\,\text{mm} = 0.45\,\text{mm}\downarrow$$

$$\sigma_A = \frac{50\times10^3}{140}\,\text{MPa} = 357.14\,\text{MPa}$$

$$357.14 = \frac{124000\varepsilon}{1+240\varepsilon} \qquad \varepsilon = 9.328\times10^{-3}$$

$$\delta_A = 2\times9.328\times10^{-3}\,\text{m} = 18.656\,\text{mm}$$

（4） 變形分析



由位移圖 $\delta_C - \delta_B = 2(\delta_A + \delta_B)$

$$\delta_C = 2\delta_A + 3\delta_B = 38.662\,\text{mm}\downarrow$$

（5） 卸載分析

卸載後 BE 桿件回復未變形 $\delta_B = 0$

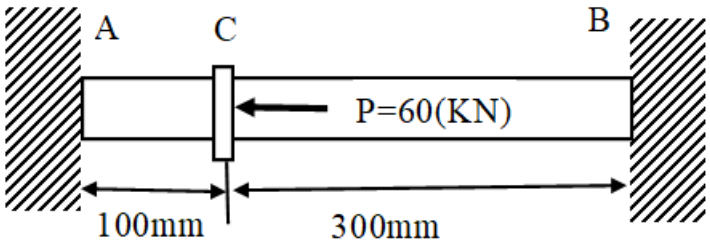
計算 AD 桿件之殘留

$$\varepsilon_P = \varepsilon - \frac{\sigma}{E} = 9.328\times10^{-3} - \frac{357.14\,\text{MPa}}{124\,\text{GPa}} = 6.448\times10^{-3}$$

$$\Delta L_P = L\times\varepsilon_p = 2\times6.448\times10^{-3}\,\text{m} = 12.896\,\text{mm}$$

例 4

A 20 mm diameter bar AB has a circle cross section and is made of a steel that is assumed to be elastoplastic with $E = 70(GPa)$ and $\sigma_y = 420(MPa)$. Knowing that the force P increases from 0 to 60 kN and then decreases to zero, determine (a) the permanent deflection of point C, (b) the residual stress in the bar



【中文翻譯】

如圖所示一靜不定桿件之桿直徑為10mm且由彈-完全塑性材料製成，該桿之彈性模數 $E = 70(GPa)$、降伏強度 $\sigma_y = 420(MPa)$，若一力P從0到60KN作用在桿子上而後移去，試求桿中之殘留應力及軸環C之永久位移

【解析】

(一)內力分析

$F_A = 45(KN)$ 、 $F_B = 15(KN)$

(二)應力分析

AC段應力分析

$$\sigma_{AC} = \frac{45\text{ kN}}{\pi(0.005\text{ m})^2} = 537\text{ MPa}\ (\text{壓縮}) > \sigma_Y = 420\text{ MPa}$$

BC段應力分析

$$\sigma_{CB} = \frac{15\text{ kN}}{\pi(0.005\text{ m})^2} = 191\text{ MPa}\ (\text{拉伸})$$

分析後AC段降伏

(三)AC段降伏重新應力與變形分析

AC段降伏則 F_A 作用力重新計算

$$(F_A)_Y = \sigma_Y A = 420(10^3)\text{ kN/m}^2[\pi(0.005\text{ m})^2]$$
$$= 33.0\text{ kN}$$

由靜力平衡

$$F_B = 60 - 33 = 27(KN)$$

$$\sigma_{AC} = \sigma_Y = 420 \text{ MPa (壓)}$$

$$\sigma_{CB} = \frac{27.0 \text{ kN}}{\pi(0.005 \text{ m})^2} = 344 \text{ MPa (拉)} < 420 \text{ MPa (OK)}$$

(四)殘留應力

由BC段彈性分析

$$\delta_C = \frac{F_B L_{CB}}{EA} = \frac{27 \times 0.3}{\pi \times 0.005^2 \times 70 \times 10^6} = 0.001474(m)$$

$$\varepsilon_{BC} = \frac{\delta_C}{L_{BC}} = \frac{0.001474}{0.3} = 0.004913$$

AC的變形一樣為δ_C

$$\varepsilon_{AC} = \frac{\delta_C}{L_{AC}} = \frac{0.001474}{0.1} = -0.01474$$

當P作用後卸載則一反向力$F_B = 15(KN)$及$F_A = 45(KN)$作用在各段

這些力所產生之應力$\sigma_{AC} = 573(MPa)$及$\sigma_{BC} = -191(MPa)$

殘留應力

$$(\sigma_{AC})_r = -420 + 573 = 153(MPa)$$

$$(\sigma_{BC})_r = -344 - 191 = 153(MPa)$$

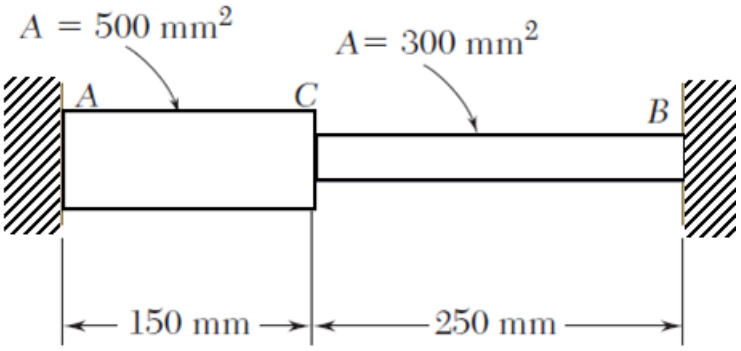
(五)永久變形

$$(\varepsilon_{BC})_r = \frac{\sigma}{E} = \frac{153 \times 10^6}{70 \times 10^9} = 0.002185$$

$$\delta_C = (\varepsilon_{BC})_r L_{BC} = 0.002185 \times 300(mm) = 0.656mm$$

例 5

The steel rod ABC is attached to rigid supports and is unstressed at a temperature of 25℃. The steel is assumed elastoplastic with E=200 GPa and $\sigma_y = 250\text{MPa},$. The temperature of both portions of the rod is then raised to 150℃. Knowing that $\alpha = 11.7\times10^{-6}/^{\circ}C$, determine (a) the stress in both portions of the rod, (b) the deflection of point C.



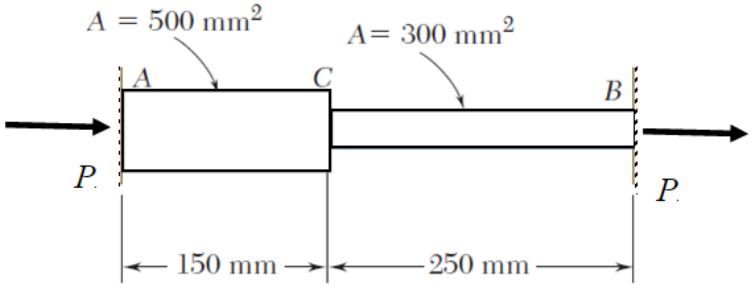
【中山機械】

【中文翻譯】

一鋼桿ABC連接在剛性支架上，在25℃的溫度下不受力。 假定鋼材為彈塑性材料，E = 200 GPa 和 $\sigma_y = 250\text{MPa}$ ，若將溫度然後升高到150℃，已知 $\alpha = 11.7\times10^{-6}/^{\circ}C$ ，試求（a）桿兩部分的應力（b）C點的變形量

【解析】

(一)內力分析



$A_{AC} = 500\times10^{-6}\text{m}^2 \qquad L_{AC} = 0.150\text{ m}$

$A_{CB} = 300\times10^{-6}\text{m}^2 \qquad L_{CB} = 0.250\text{ m}$

(二)變形分析

$$\alpha\Delta TL=\frac{PL_{AC}}{EA_{AC}}+\frac{PL_{BC}}{EA_{BC}}$$

$$(11.7\times10^{-6})(125)(400)=\frac{P(150)}{(200\times10^{3})(500)}+\frac{P(250)}{(200\times10^{3})(300)}$$

$$P=103.23(\text{kN})$$

(三)BC段應力

$$\sigma_{BC}=\frac{P}{A_{BC}}=344.1(\text{MPa})>\sigma_{yp}$$

所以 BC 段已降伏$\Rightarrow\sigma_{BC}=\sigma_{yp}=250(\text{MPa})$（壓應力）

$$P=\sigma_{BC}A_{BC}=75(\text{kN})$$

(四)AC段應力

$$P=P_Y=75\times10^{3}\,\text{N}$$

$$\sigma_{AC}=-\frac{P_Y}{A_{AC}}=-\frac{75\times10^{3}}{500\times10^{-6}}=-150\times10^{-6}\,\text{Pa}\qquad\sigma_{AC}=-150.0\ \text{MPa}$$

$$\sigma_{CB}=-\frac{P_Y}{A_{CB}}=-\sigma_Y\qquad\sigma_{CB}=-250\ \text{MPa}$$

(五)變位計算

$$\delta_{C/A}=-\frac{P_YL_{AC}}{EA_{AC}}+L_{AC}\alpha(\Delta T)$$

$$=-\frac{(75\times10^{3})(0.150)}{(200\times10^{9})(500\times10^{-6})}+(0.150)(11.7\times10^{-6})(125)=106.9\times10^{-6}\,\text{m}$$

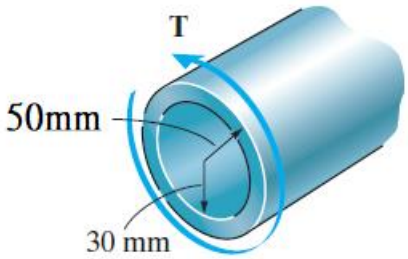
$$\delta_C=\delta_{C/A}=106.9\times10^{-6}\,\text{m}$$

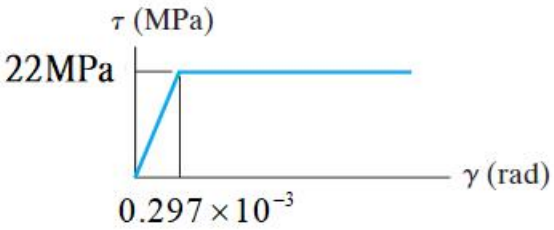
例 6

ln Figure 3 the aluminum alloy tubu1ar shaft is assumed to have an elastic-plastic $\tau-\gamma$ relation as shown.

(a) Determine the maximum torque that can be applied to the shaft without causing the material to yield. (8%)

(b) Find the maximum torque (or plastic torque) that can be applied to the shaft





【103 台大應力】

【解析】

(一)斷面分析

$$J=\frac{\pi}{2}\left(50^4-30^4\right)=8.545\times10^6\ \text{mm}^4$$

(二)扭矩分析

$$\tau_y=\frac{T_y r}{J}\Rightarrow 22=\frac{T_y\times 50}{8.545\times10^6}\Rightarrow T_y=3759800(N-mm)=3759.8(N-m)$$

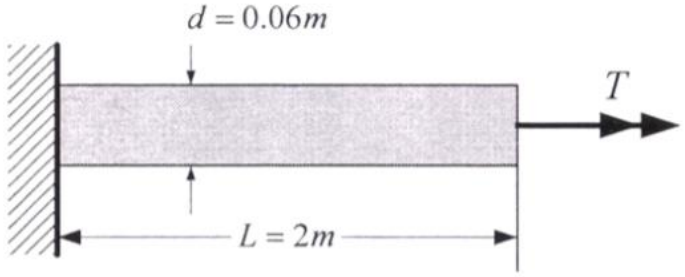
$$T_P=2\pi\int\tau\rho^2 d\rho=2\pi\int_{30}^{50}\tau_y\rho^2 d\rho\Rightarrow T_P=4515500\ (N-mm)=4515.5(N-m)$$

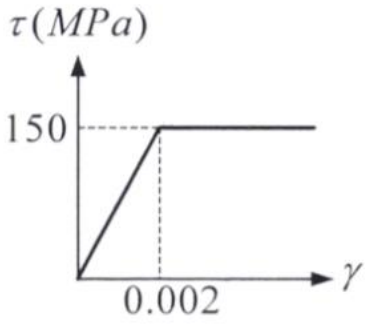
例 7

長 $L=2\ m$，半徑 $R=0.03\ m$ 之實心圓桿受到扭矩 T 作用，如圖 1 所示。此實心圓桿為理想塑性材料（perfect plasticity），求：

㈠此桿之最大彈性扭矩 T_y（即此桿剛剛降伏時之扭矩）。（5 分）

㈡當 $T=1.2\ T_y$ 時卸載，求卸載後此圓桿之殘留應力分布狀況。（15 分）





【土木三等】

【解析】

(一)扭矩分析

$$\tau_y=\frac{T_y r}{J}\Rightarrow 150\times10^6=\frac{T_y\times0.03}{\frac{\pi}{2}\times0.03^4}\Rightarrow T_y=6362(N-m)$$

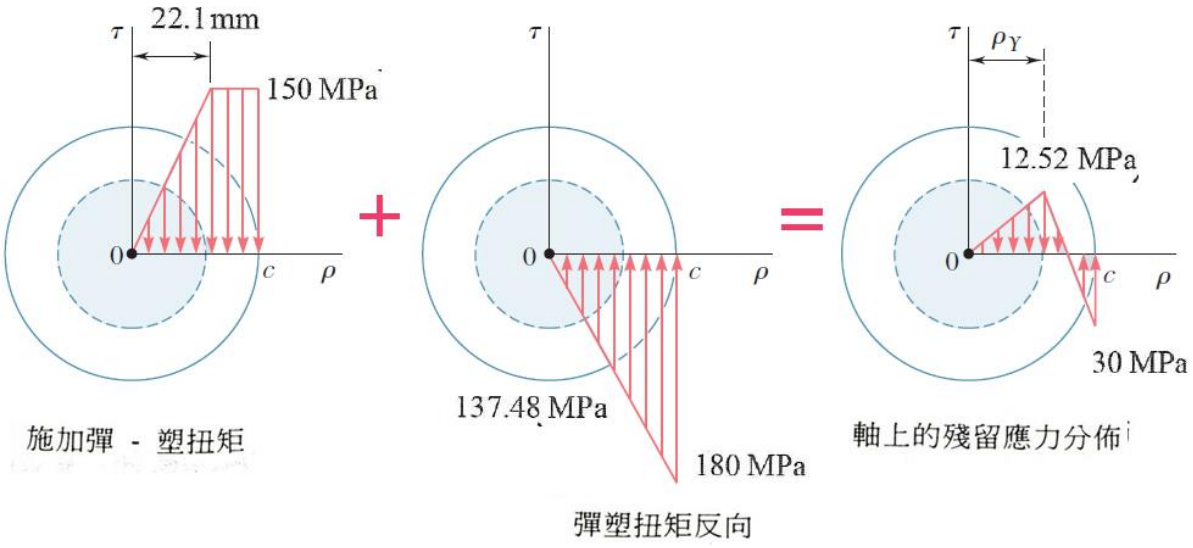
$$T=\frac{4}{3}T_y\left(1-\frac{\rho_y^3}{4r^3}\right)=1.2T_y\ ,\ \rho_y=0.737r$$

$$\Rightarrow \rho_y=0.0221(m)=22.1(mm)$$

(二)將扭矩移除

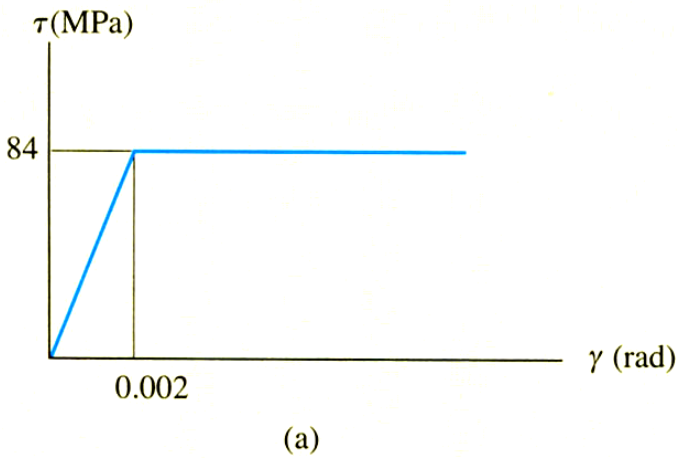
彈性回復之扭轉角 ϕ' 及剪應力 τ'

$$\tau'=\frac{Tr}{J}=\frac{6362\times1.2\times0.03}{\frac{\pi}{2}\times0.03^4}=180\times10^6(pa)=180(MPa)$$


τ
22.1 mm
150 MPa
0
c
ρ
施加彈 - 塑扭矩
+
τ
0
c
ρ
137.48 MPa
180 MPa
彈塑扭矩反向
=
τ
ρY
12.52 MPa
0
c
ρ
30 MPa
軸上的殘留應力分佈

例 8

一彈塑材料(τ_Y = 84 MPa，G = 42 GPa)製造之空心圓桿，內徑為 50mm，外徑為 100mm，長度為 1.5m。試求(a)使圓桿斷面完全降伏所需之扭矩(塑性扭矩 T_P)？(b)將(a)中之扭矩移去後，試求永久扭轉角及殘留應力。



【解析】

(一)扭矩分析

塑性扭矩 $\mathbf{T}_p$ 將使管子應變至所有材料均降伏

$$T_p = 2\pi \int_{c_i}^{c_o} \tau_Y \rho^2 d\rho = \frac{2\pi}{3}\tau_Y (c_o^3 - c_i^3)$$

$$= \frac{2\pi}{3}(84\ \text{N/mm}^2)[(50\ \text{mm})^3 - (25\ \text{mm})^3]$$

$$= 19.24(10^6)\ \text{N}\cdot\text{mm}$$

$$\phi_p = \gamma_Y \frac{L}{c_i} = \frac{(0.002)(1.5\ \text{m})(10^3\ \text{mm/m})}{(25\ \text{mm})} = 0.120\ \text{rad}\curvearrowleft$$

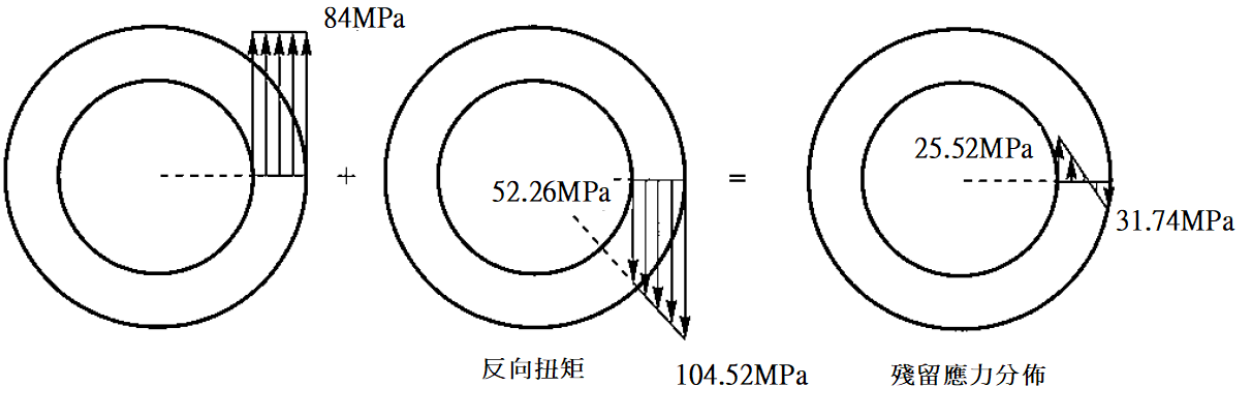
(二)將扭矩移除

將扭矩移除，彈性回復之扭轉角 ϕ' 及剪應力 τ_r 為

$$\tau_r = \frac{T_p c_o}{J} = \frac{19.24(10^6)\ \text{N}\cdot\text{mm}\,(50\ \text{mm})}{(\pi/2)[(50\ \text{mm})^4 - (25\ \text{mm})^4]}$$

$$= 104.52\ \text{N/mm}^2 = 104.52\ \text{MPa}$$

$$\tau_i = (104.52\ \text{MPa})\left(\frac{25\ \text{mm}}{50\ \text{mm}}\right) = 52.26\ \text{MPa}$$



$$\phi'_p = \frac{T_p L}{JG}$$

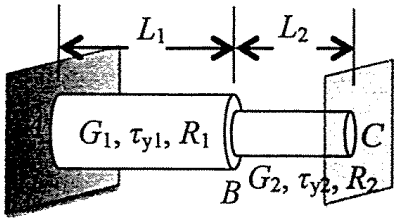
$$= \frac{19.24(10^6)\,\text{N}\cdot\text{mm}\,(1.5\ \text{m})(10^3\,\text{mm/m})}{(\pi/2)[(50\ \text{mm})^4 - (25\ \text{mm})^4]42\,(10^3)\text{N/mm}^2}$$

$$= 0.0747\ \text{rad}\ \curvearrowright$$

$$\phi = 0.120 - 0.0747 = 0.0453\ \text{rad}\curvearrowleft$$

例 9

兩根實心圓柱在 B 點黏合，並在 A 點與 C 點固定於牆面上；兩根圓柱均為線彈性-完美塑性材料，G_1、G_2 分別為其剪力模數，τ_{y1}、τ_{y2} 分別為其降伏剪應力，R_1、R_2 分別為其半徑，L_1、L_2 分別為其長度；整體結構在 B 點承受一個軸向扭矩 T。請推導並回答以下問題。



(1) 圓柱 1 優先於圓柱 2 降伏的條件是什麼？
請以 G_1、G_2、τ_{y1}、τ_{y2}、R_1、R_2、L_1、L_2、T 表示。(10 分)

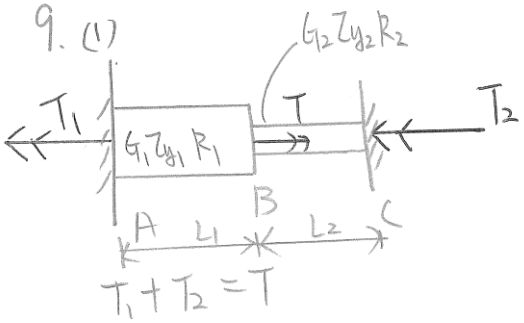
(2) 若圓柱 1 先於圓柱 2 降伏，當圓柱 2 開始降伏時，扭矩 T 為何? (10 分)

(3) 若圓柱 1 先於圓柱 2 降伏，當圓柱 2 開始降伏時，B 點的扭轉角度為何？(10 分)

【105 台大機械】

【解析】

9. (1)



$T_1 + T_2 = T$

由變位一致性

$$\phi_{BC} = \frac{T_2 \times L_2}{G_2 J_2}$$

$$\phi_{AB} = \frac{T_1 \times L_1}{G_1 J_1}$$

$$\phi_{AB} = \phi_{BC} \Rightarrow \frac{T_1}{T_2} = \frac{L_2 G_1 J_1}{L_1 G_2 J_2}$$

$$\tau_1 = \frac{T_1 R_1}{J_1} = \frac{2T_1}{\pi R_1^3}$$

$$\tau_2 = \frac{T_2 R_2}{J_2} = \frac{2T_2}{\pi R_2^3}$$

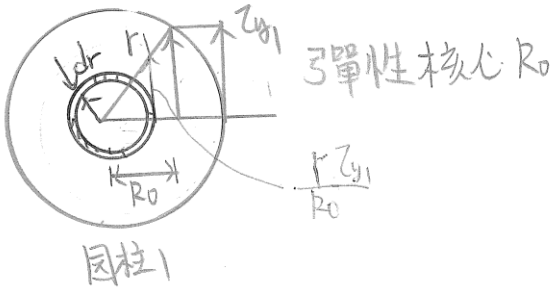
$$\frac{\tau_1}{\tau_2} = \frac{T_1}{T_2} \times \frac{R_2^3}{R_1^3} = \frac{L_2 G_1 R_1}{L_1 G_2 R_2}$$

若圓柱1先達到降伏 $\Rightarrow \tau_2 < \tau_{y2}$, $\tau_1 = \tau_{y1}$

$$\frac{\tau_1}{\tau_2} > \frac{\tau_{y1}}{\tau_{y2}}$$

$$\therefore \frac{L_2 G_1 R_1}{L_1 G_2 R_2} > \frac{\tau_{y1}}{\tau_{y2}}$$

(2) 圓柱1已降伏 → 圓柱2降伏



$T_1 = (T_1)_{\text{彈性區}} + (T_1)_{\text{塑性區}}$

$$= \int_0^{R_0} \left(\frac{r}{R_0}\tau_{y1}\right) \cdot r \cdot (2\pi r\, dr) + \int_{R_0}^{R_1} \tau_{y1} \cdot r \cdot (2\pi r\, dr)$$

$$= \underline{\frac{1}{6}\pi \tau_{y1}(4R_1^3 - R_0^3)} \ \text{※}$$

圓柱 2 開始降伏

$$\tau_2 = \tau_{y2} = \frac{T_2 R_2}{J_2} \Rightarrow T_2 = \frac{\tau_{y2} \pi R_2^3}{2} \quad \text{※}$$

$$\phi_B = \frac{T_2 L_2}{G_2 J_2} = \frac{\tau_{y2} L_2}{G_2 R_2} \quad \text{※} \quad (\text{第(3)題解答})$$

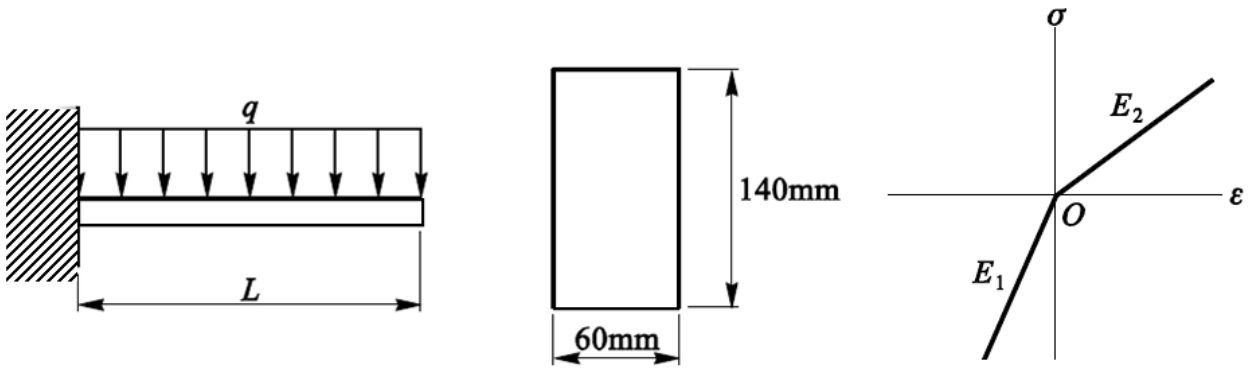
$$\gamma = R\theta = R_0 \times \frac{\phi_B}{L_1} = \frac{\tau_{y1}}{G_1}$$

$$\Rightarrow R_0 = \frac{\tau_{y1} \times L_1}{\phi_B \times G_1} = \frac{\tau_{y1} L_1 G_2 R_2}{G_1 \tau_{y2} L_2} \quad \text{代入 } T_1$$

故 $T = T_1 + T_2$

例 10

圖中長度 L = 1.0m 之懸臂梁承受均佈負荷 q = 12.0 kN/m。材料拉伸與壓縮之彈性模數分別為 E_2 = 70 GPa 與 E_1 = 200GPa，試求梁內所生之最大拉應力與最大壓應力。



【成大機械】

【解析】

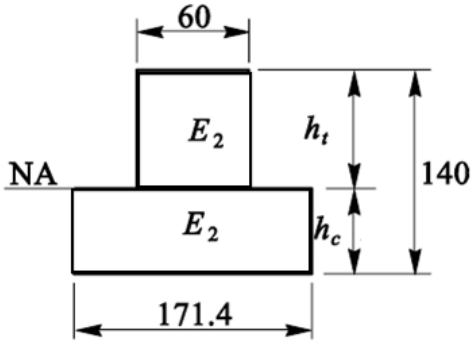
(一)斷面分析

將斷面轉為 E_2，$n=\dfrac{E_1}{E_2}=\dfrac{200}{70}$，下緣之寬度轉為 $\dfrac{200}{70}\times 60=171.4$ mm

斷面對中立軸之一次矩為零

$$(171.4h_c)\left(\frac{h_c}{2}\right)=60(140-h_c)\left(\frac{140-h_c}{2}\right)\text{，}h_c=52\text{ mm}$$

$$h_t=140-h_c=88\text{ mm}\text{，}I=\frac{1}{3}(60)(88)^3+\frac{1}{3}(171.4)(52)^3=21.66\times10^6\text{ mm}^4$$



(二) 應力分析

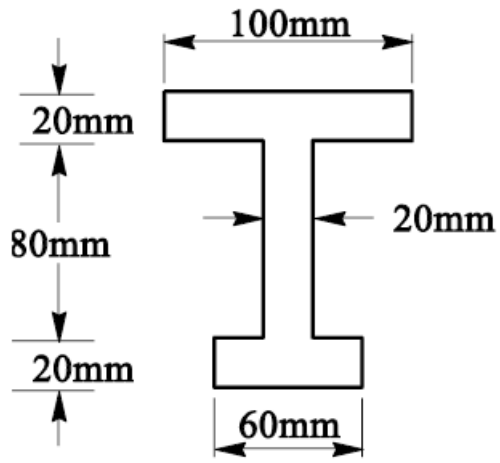
梁內最大彎矩：$M=\dfrac{1}{2}qL^2=\dfrac{1}{2}(12.0)(1)^2=6.0$ kN-m

$$\sigma_t=\frac{Mh_t}{I}=\frac{(6.0\times10^6)(88)}{21.66\times10^6}=24.4\text{ MPa}$$

$$\sigma_c=n\frac{Mh_c}{I}=\left(\frac{200}{70}\right)\frac{(6.0\times10^6)(52)}{21.66\times10^3}=41.2\text{ MPa}$$

例 11

Determine the plastic moment Mp of a beam with the cross section shown when the beam is bent about a horizontal axis. Assume that the material is elastoplastic with a yield strength of $\sigma_y = 240(MPa)$.

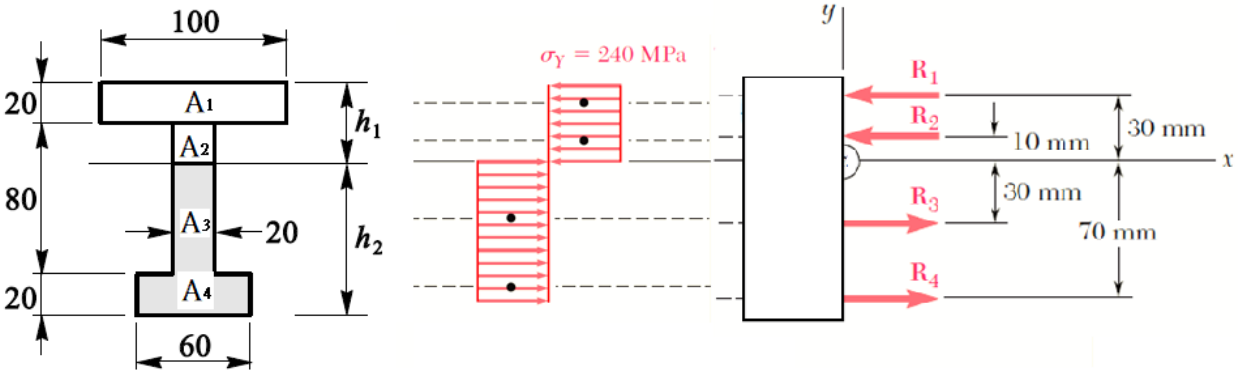


【解析】

圖中所示斷面之梁為彈塑材料所製造，降伏應力為 $\sigma_y = 240(MPa)$，試求梁之塑性彎矩Mp

(一)塑性斷面分析

$$R_1 = A_1\sigma_Y = [(0.100\text{ m})(0.020\text{ m})]240\text{ MPa} = 480\text{ kN}$$
$$R_2 = A_2\sigma_Y = [(0.020\text{ m})(0.020\text{ m})]240\text{ MPa} = 96\text{ kN}$$
$$R_3 = A_3\sigma_Y = [(0.020\text{ m})(0.060\text{ m})]240\text{ MPa} = 288\text{ kN}$$
$$R_4 = A_4\sigma_Y = [(0.060\text{ m})(0.020\text{ m})]240\text{ MPa} = 288\text{ kN}$$

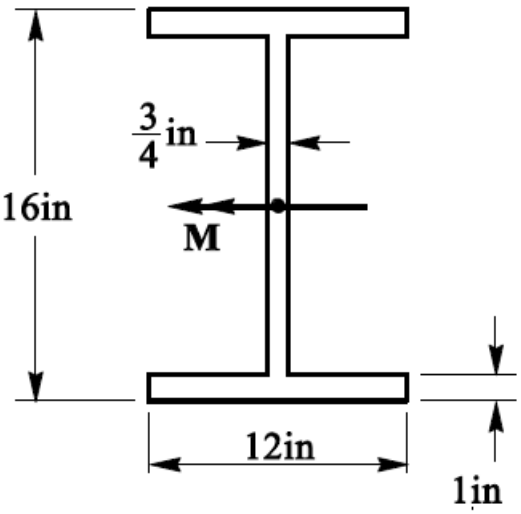


(二)塑性分析

$$
\begin{aligned}
M_p &= (0.030\ \text{m})R_1 + (0.010\ \text{m})R_2 + (0.030\ \text{m})R_3 + (0.070\ \text{m})R_4 \\
&= (0.030\ \text{m})(480\ \text{kN}) + (0.010\ \text{m})(96\ \text{kN}) \\
&\quad + (0.030\ \text{m})(288\ \text{kN}) + (0.070\ \text{m})(288\ \text{kN}) \\
&= 44.16\ \text{kN} \cdot \text{m}
\end{aligned}
$$

例 12

Beam AB has been fabricated from a high-strength low-alloy steel that is assumed to be elastoplastic with $E = 29\times10^6(psi)$ and $\sigma_y = 50(ksi)$. Neglecting the effect of fillets, (a) determine the bending moment M and the corresponding radius of curvature when yield first occurs, when the flanges have just become fully plastic.(b) For the beam of Sample (a)determine the residual stresses and the permanent radius of curvature after the couple M has been removed.



【中文翻譯】

樑AB由高強度低合金鋼製成，假定其為彈塑性材料($\sigma_y = 50(ksi)$，$E = 30(ksi)$)製造， 忽略圓角的影響，(a)試求使上下翼板完全降伏所需之彎矩M及此時梁之曲率半徑，(b)將(a)中之彎矩M 移去，試求斷面之殘留應力及永久變形之曲率半徑。

【解析】

(a)

(一)剛開始降伏時斷面的形心慣性矩

$$I = \tfrac{1}{12}(12\text{ in.})(16\text{ in.})^3 - \tfrac{1}{12}(12\text{ in.} - 0.75\text{ in.})(14\text{ in.})^3 = 1524\text{ in}^4$$

降伏彎矩

$$\mathrm{M_y} = \frac{\sigma_y \mathrm{I}}{y} = \frac{50\times1524}{8} = 9525(Kip-in)$$

(二)曲率半徑

$$\varepsilon_y = \frac{\sigma_y}{E} = \frac{50\times10^3}{29\times10^6} = 0.001724$$

$$\rho = \frac{y_Y}{\varepsilon_y} = \frac{8}{0.001724} = 4640(in)$$

(三)上下翼板完全降伏

合力R_1、R_2為壓應力之合力，同理R_3、R_4為拉應力之合力

$$R_1 = R_4 = (50\ \text{ksi})(12\ \text{in.})(1\ \text{in.}) = 600\ \text{kips}$$
$$R_2 = R_3 = \tfrac{1}{2}(50\ \text{ksi})(7\ \text{in.})(0.75\ \text{in.}) = 131.3\ \text{kips}$$

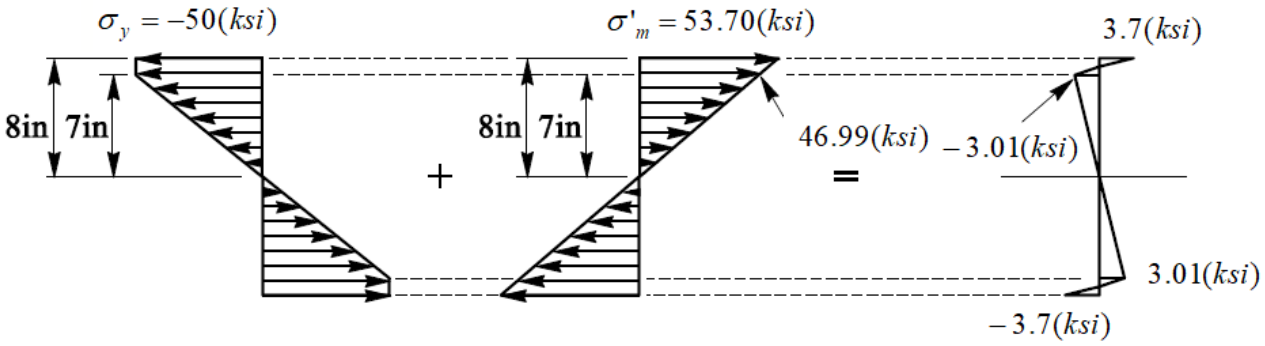
$$M = 2[R_1(7.5\ \text{in.}) + R_2(4.67\ \text{in.})]$$
$$= 2[(600)(7.5) + (131.3)(4.67)] \qquad M = 10{,}230\ \text{kip} \cdot \text{in.}$$

$$\rho = \frac{y_Y}{\varepsilon_y} = \frac{7}{0.001724} = 4060(in)$$

(b)殘留應力

彎矩移去後產生彈性回復，相當於施加一反向彎矩M' = 10230(Kip-in) 抵銷原施加之彎矩

$$\sigma'_m = \frac{Mc}{I} = \frac{(10{,}230\ \text{kip} \cdot \text{in.})(8\ \text{in.})}{1524\ \text{in}^4} = 53.70\ \text{ksi}$$



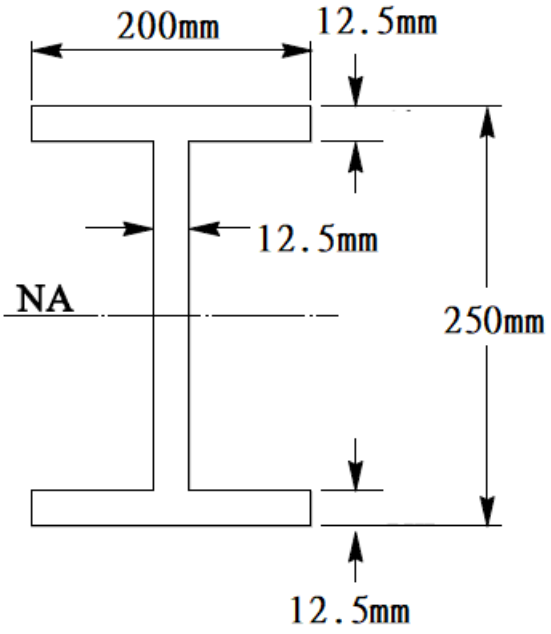
曲率半徑

$$\varepsilon_x = \frac{\sigma_x}{E} = \frac{-3.01\times10^3}{29\times10^6} = -1.038\times10^{-4}$$

$$\rho = \frac{y_Y}{\varepsilon_x} = -\frac{7}{-1.038\times10^{-4}} = 67441.86(in)$$

例 13

如圖所示一 I 型雙對稱斷面之寬翼鋼樑，受到其強軸之彎矩作用，已知該材料為理想的彈塑性材料，具拉伸及壓縮降伏應力 $\sigma_y = 250(MPa)$，試求(一)樑之形狀因數 f(二)若將塑性彎矩釋放試求殘留應力分佈



【機械技師】

【解析】

(二)形狀因子

(a)斷面分析

$$I = [\frac{1}{12} \times 200 \times 250^3] - [\frac{1}{12} \times 187.5 \times 225^3] = 82.44 \times 10^6 (mm^4)$$

(b)應力分析

$$\sigma_y = \frac{M_y y}{I} \Rightarrow 250 = \frac{M_y \times 125}{82.44 \times 10^6} \Rightarrow M_y = 164880000(N-mm) = 164.88(KN-m)$$

(c)塑性彎矩

塑性彎矩使樑整個截面上產生降伏，故計算截面上映力分佈之體積，將應力分佈區分為兩區

$$C_1 = T_1 = 250\ \text{N/mm}^2(12.5\ \text{mm})(112.5\ \text{mm}) = 351.56\ \text{kN}$$
$$C_2 = T_2 = 250\ \text{N/mm}^2(12.5\ \text{mm})(200\ \text{mm}) = 625\ \text{kN}$$

$$M_p = 2[(56.25\ \text{mm})(351.56\ \text{kN})] + 2[(118.75\ \text{mm})(625\ \text{kN})] = 188\ \text{kN}\cdot\text{m}$$

形狀因子 $= \dfrac{M_p}{M_y} = \dfrac{188}{164.8} = 1.14$

(三)塑性彎矩釋放

塑性彎矩釋放後，需要一反向 $M_P = 188(KN-m)$

$$\sigma_r = \frac{M_p y}{I} = \frac{188\times10^6\times125}{82.44\times10^6} = 285.1(MPa)$$

殘留應力如下所示

